

Git Hands-On Lab

Lab 1: Git Setup, Configuration & First Commit

```
git --version
```

```
git config --global user.name "Your Name"  
git config --global user.email "youremail@example.com"
```

```
git config --global --list
```

```
notepad++
```

```
notepad++ ~/.bashrc
```

```
alias notepad++="/c/Program Files/Notepad++/notepad++.exe"  
source ~/.bashrc
```

```
git config --global core.editor "'C:/Program Files/Notepad++/notepad++.exe' -multiInst -  
notabbar -nosession -noPlugin"
```

```
git config -e
```

```
mkdir GitDemo && cd GitDemo  
git init
```

```
ls -la
```

```
echo "Welcome to GitDemo project" > welcome.txt
```

```
ls
```

```
cat welcome.txt
```

```
git status
```

```
git add welcome.txt
```

```
git commit
```

```
git status
```

```
git remote add origin https://gitlab.com/<your-username>/GitDemo.git  
git pull origin master --allow-unrelated-histories  
git push origin master
```

Lab 2: Git Ignore

```
echo "sample log entry" > app.log  
mkdir log  
echo "another log entry" > log/debug.log
```

```
git status
```

```
notepad++ .gitignore
```

```
*.log  
log/
```

```
git status
```

```
git add .gitignore  
git commit -m "Add .gitignore to exclude log files and log folder"
```

```
git status  
git push origin master
```

Lab 3: Branching and Merging

```
git branch GitNewBranch
```

```
git branch -a
```

```
git checkout GitNewBranch  
echo "Content added on GitNewBranch" > feature.txt
```

```
git add feature.txt  
git commit -m "Add feature.txt on GitNewBranch"
```

```
git status
```

```
git checkout master
```

```
git diff master GitNewBranch
```

```
git difftool -t p4merge master GitNewBranch
```

```
git merge GitNewBranch
```

```
git log --oneline --graph --decorate
```

```
git branch -d GitNewBranch  
git branch  
git status
```

Lab 4: Merge Conflict Resolution

```
git checkout master
git status
```

```
git checkout -b GitWork
echo "<greeting>Hello from GitWork</greeting>" > hello.xml
git add hello.xml
git commit -m "Add hello.xml on GitWork"
```

```
echo "<greeting>Hello from GitWork - updated</greeting>" > hello.xml
git status
git add hello.xml
git commit -m "Update hello.xml content on GitWork"
```

```
git checkout master
```

```
echo "<greeting>Hello from master</greeting>" > hello.xml
git add hello.xml
git commit -m "Add hello.xml on master"
```

```
git log --oneline --graph --decorate --all
```

```
git diff master GitWork -- hello.xml
```

```
git difftool -t p4merge master GitWork -- hello.xml
```

```
git merge GitWork
```

```
git mergetool -t p4merge
```

```
git add hello.xml
git commit -m "Merge GitWork into master, resolve hello.xml conflict"
```

```
git status
```

```
notepad++ .gitignore
```

```
*.orig
```

```
git add .gitignore
git commit -m "Ignore .orig backup files created by mergetool"
```

```
git branch
```

```
git branch -d GitWork
```

```
git log --oneline --graph --decorate
```


Lab 5: Clean Up and Push Back to Remote

```
git checkout master  
git status
```

```
git branch -a
```

```
git pull origin master
```

```
git push origin master
```

```
git log origin/master --oneline -5
```